

TextFX: Reconstructing Editable Stylized Typography Through Parametric Text Effect Extraction

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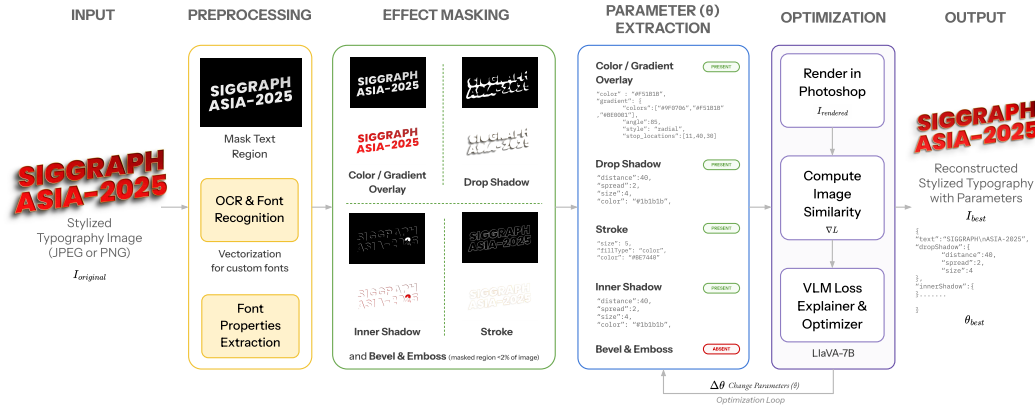
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Graphical Abstract: Reverse Engineering of Text Effects at a Glance

Abstract

Stylized text plays a crucial role in graphic design, digital art, and media, yet most existing stylizations are embedded in flattened raster images, limiting editability and reuse. This paper presents *TextFX*, a framework that reconstructs stylized text from raster images into editable, parameterized layers. TextFX decomposes complex visual effects—such as shadows, gradients, strokes, and embossing—into interpretable style parameters. Quantitative results show consistent improvement across image similarity metrics (SSIM, RMSE, SRE, SAnM), and user studies with designers and novices confirm high usability, quality, and adoption potential. Experiments also showed that the addition of human-in-the-loop corrections during refinement, substantially improves the accuracy and perceptual quality of the reconstruction. TextFX, therefore, bridges the gap between rasterized artwork and structured design, enabling flexible, editable typography and setting the stage for prompt-driven, generative design workflows.

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1 Introduction

Stylized text is a key component in graphic design, advertising, and movie titles, where visual form enhances meaning. Despite rising demand for creative AI tools that support editability and reuse, most real-world stylized text exists as flattened raster images, making direct manipulation impossible. Designers seeking to edit or repurpose such text must manually recreate effects in software like Photoshop or Illustrator [Avudaiappan and Murali 2025]. Recent progress in stylized text generation has ranged from patch-based synthesis in Awesome Typography [Yang et al. 2016], to style transfer [Sai et al. 2025], to GAN-based methods such as TET-GAN [Yang et al. 2019], and more recent diffusion-driven models like TextCtrl [Zeng et al. 2025].

While these methods improve realism, they remain limited by poor integration with professional design tools, producing flat, non-editable images. To bridge this gap, we introduce a framework that reconstructs stylized text effects from raster images into editable,

parameterized layers, enabling flexible control over typography design.

2 Methodology

The *TextFX* framework reconstructs stylized text effects from raster images into parameterized, editable layers. The pipeline consists of five sequential stages: preprocessing, effect masking, parameter extraction, optimization, and output reconstruction. The system takes as input a stylized typography image $I_{original}$, in either JPEG or PNG format. Such images typically contain layered effects (e.g., gradient overlays, shadows, strokes, embossing) that are visually entangled with the text content, making decomposition non-trivial.

Preprocessing begins with text region segmentation, followed by OCR via Tesseract and font classification using a fine-tuned EfficientNet-B3 trained on Google Fonts, with vectorization for unseen fonts. Font attributes—color (k-means), alignment (character centers), and size (median height)—are then extracted to form a symbolic-stylistic representation for effect decomposition. The **Effect Masking** stage isolates visual effects from the base text using targeted computer vision methods. Specific heuristics are applied per effect: color and gradient overlays via intensity and gradient analysis; shadows and glows via morphological differencing and distance transforms; strokes through contour expansion and subtraction; and bevel/emboss effects using gradient and illumination cues. This process produces binary masks for each stylistic layer. In the **Parameter Extraction** stage, each masked effect is compared against the original image to estimate key attributes. Extracted parameters include spatial, color, and shape properties—e.g., shadow offset and blur, stroke width and color, gradient stops and angles, or emboss elevation and light intensity. Each effect is also explicitly tagged, producing a structured set of parameters (θ) per text instance. Finally, in the **Optimization** stage, parameters are rendered in Photoshop to produce $I_{rendered}$, which is compared with $I_{original}$ using RMSE, SRE, SAnM, and SSIM. A Vision-Language Model (LLaVA-7B) interprets discrepancies and suggests updates using the triplet ($I_{original}$, $I_{rendered}$, loss). Refinement continues iteratively until convergence, with stopping criteria defined as $SSIM \geq 0.95$, $RMSE \leq 0.05$, or manual confirmation. The final output, I_{best} , is both visually accurate and fully parameterized. Alongside the rendered image, the system outputs a structured, machine-readable representation (JSON-like) of typographic effects, enabling editable



(a) Rendered image alongside original input

Experiment	RMSE	SRE (dB)	SAnM (°)	SSIM
1 iter	0.085	18.2	8.6	0.912
5 iters	0.060	22.4	5.3	0.952
1 iter + Edit	0.055	23.1	4.9	0.957
5 iters + Edit	0.035	26.7	3.1	0.975

(b) Similarity metrics across different experiments

Figure 1: Comparison between rendered outputs and similarity scores for various reconstruction iterations.

and reusable typography. This bridges raster visuals and structured design data for downstream editing or style transfer.

3 Results & User Feedback

The initial framework design included only a single-step optimization process and produced reasonable fidelity (table 1b). Additional experiments revealed that increasing the number of iterations for optimization also increased the quality of the rendered output text. For example, after 5 iterations, the RMSE, SRE, SAnM, and SSIM values obtained were a clear improvement from the single-step result. However, it is worth pointing out that the framework was able to capture the basic properties like font family, color, font size in the 1st iteration itself, as a result there was no major improvement with respect to the structural similarity (SSIM: 0.912 to 0.952). Similar experiments were also performed taking into consideration the user feedback (table 1b). Results show that having a human-in-the-loop strongly improves the optimization process towards rendering acceptable and editable text effects (figure 1a).

The framework's effectiveness was further confirmed through a user study with 35 participants, including both designers and novices. The system received consistently high ratings across five dimensions—similarity, quality, ease of use, usability, and recommendation to peers. Similarity ($\mu = 8.5$) and usability ($\mu = 9$) were particularly strong, reflecting both technical fidelity and robustness, while quality and ease ($\mu = 8$ each) showed reliable performance with minor variability. The recommendation score ($\mu = 8$) further indicates strong adoption potential, validating the framework's balance of accuracy, accessibility, and practical utility.

4 Conclusion & Future Direction

This work introduces TextFX - a structured pipeline for stylized text reconstruction with effect-aware parameter extraction. The approach significantly reduces design effort and was well received by professional users. Quantitative evaluations demonstrate strong performance, further improved through minimal manual refinement. Future work will focus on learnable parameters, differentiable rendering, and semantic embeddings to enable prompt-driven tools like *TextFX* for creative typography.

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